



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

Year 11 Course Assessment Task 3

Wednesday September 18, 2019

General instructions

- Working time – 2 hours
(plus 10 minutes reading time)
- Write your answers in the space provided.
- Write using a blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle any additional sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

☐ 11MAT.6 – Dr Lawson

NESA STUDENT #:

Marker's use only.

QUESTION	1-10	11	12	13	14	15	Total	%
MARKS	$\overline{10}$	$\overline{18}$	$\overline{10}$	$\overline{16}$	$\overline{16}$	$\overline{13}$	$\overline{83}$	

Section I

10 marks

Attempt Question 1 to 10

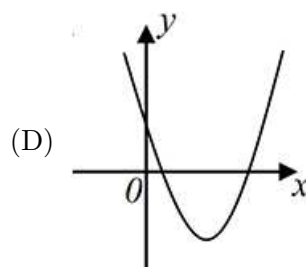
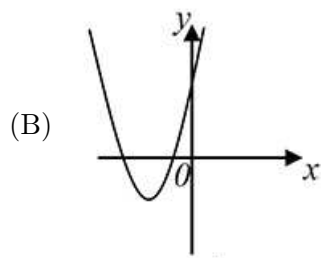
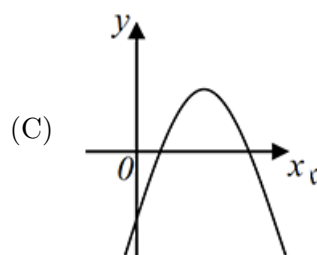
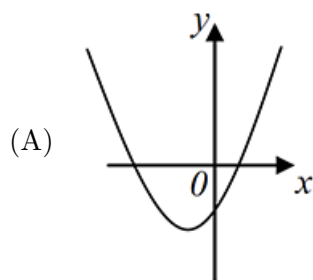
Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 18).

Questions

Marks

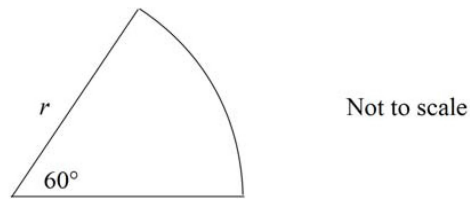
1. When the denominator is rationalised, what is $\frac{1}{\sqrt{5} - \sqrt{3}}$ equal to? 1
- (A) $\frac{\sqrt{5} - \sqrt{3}}{2}$ (B) $\frac{\sqrt{5} - \sqrt{3}}{16}$ (C) $\frac{\sqrt{5} + \sqrt{3}}{2}$ (D) $\sqrt{5} + \sqrt{3}$
2. Which of the following parabolas could have the equation $y = (x - 5)(x - 1)$? 1



3. What is range of the function $f(x) = \sqrt{4 - x^2}$? 1
- (A) $(0, 4)$ (B) $[-4, 4]$ (C) $(-2, 2)$ (D) $[0, 2]$
4. What is $e^{2 \log_e x}$ is equal to? 1
- (A) $\sqrt{x}$ (B) x^2 (C) $2x$ (D) $e^2 x$
5. If $4^{x-1} = 32$, what is the value of x ? 1
- (A) 10 (B) 3.5 (C) 3 (D) 6

6. The sector below has an area of 20π square units.

1

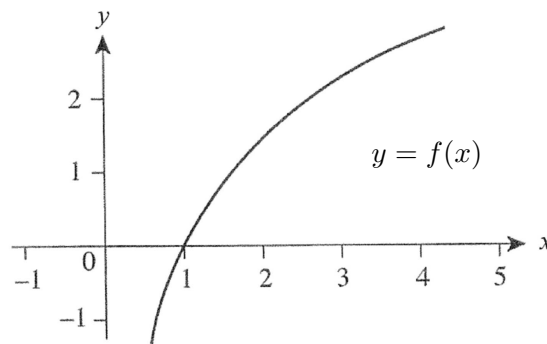


What is the value of r ?

- (A) $\sqrt{120}$ (B) $\sqrt{120}\pi$ (C) $\sqrt{\frac{\pi}{3}}$ (D) $\sqrt{\frac{1}{3}}$

7. The graph of $y = f(x)$ below has which of the following properties?

1



- (A) $f'(x) > 0$ and $f''(x) > 0$ (C) $f'(x) < 0$ and $f''(x) > 0$
 (B) $f'(x) > 0$ and $f''(x) < 0$ (D) $f'(x) < 0$ and $f''(x) < 0$

8. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$

1

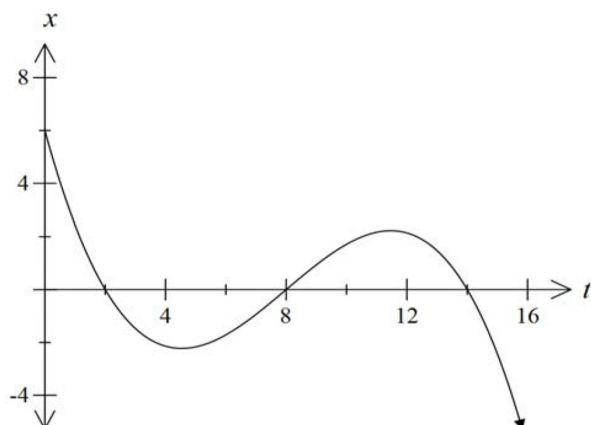
- (A) ∞ (B) 0 (C) 4 (D) 2

9. If $f(x) = a^x + a^{-x}$, which of the following expressions is equal to $[f(x)]^2$?

1

- (A) $1 - f(2x)$ (B) $1 + f(2x)$ (C) $2 - f(2x)$ (D) $2 + f(2x)$

10. The displacement, x metres, from the origin of a particle moving in a straight line at any time (t seconds) is shown in the graph. 1



When was the particle first at rest?

- (A) $t = 0$ (B) $t = 2$ (C) $t = 4.5$ (D) Never

Examination continues overleaf...

Section II

70 marks

Attempt Questions 11 to 15

Allow approximately 1 hour and 50 minutes for this section.

Question 11 (18 Marks)

Marks

- (a) Solve for x : $\frac{x+5}{3} - 2x = 7$ 2

.....

- (b) Factorise $3x^2 + 5x - 2$. 2

.....

- (c) Simplify $\frac{a^2 \times a^{x-4}}{a^{1-2x}}$ 2

.....

- (d) Consider the function $f(x) = \frac{3}{x+1}$. 1
- i. Show that $f(x)$ is neither an even nor an odd function.

.....

- ii. State the domain of $y = f(x)$ in interval notation. 1

.....

(e) Given

$$f(x) = \begin{cases} -4 & x \leq -2 \\ x & -2 < x \leq 1 \\ x^2 - 1 & x > 1 \end{cases}$$

i. Find $f(-1)$. **1**

.....

ii. Find $f(a^2 + 1)$. Simplify your answer. **2**

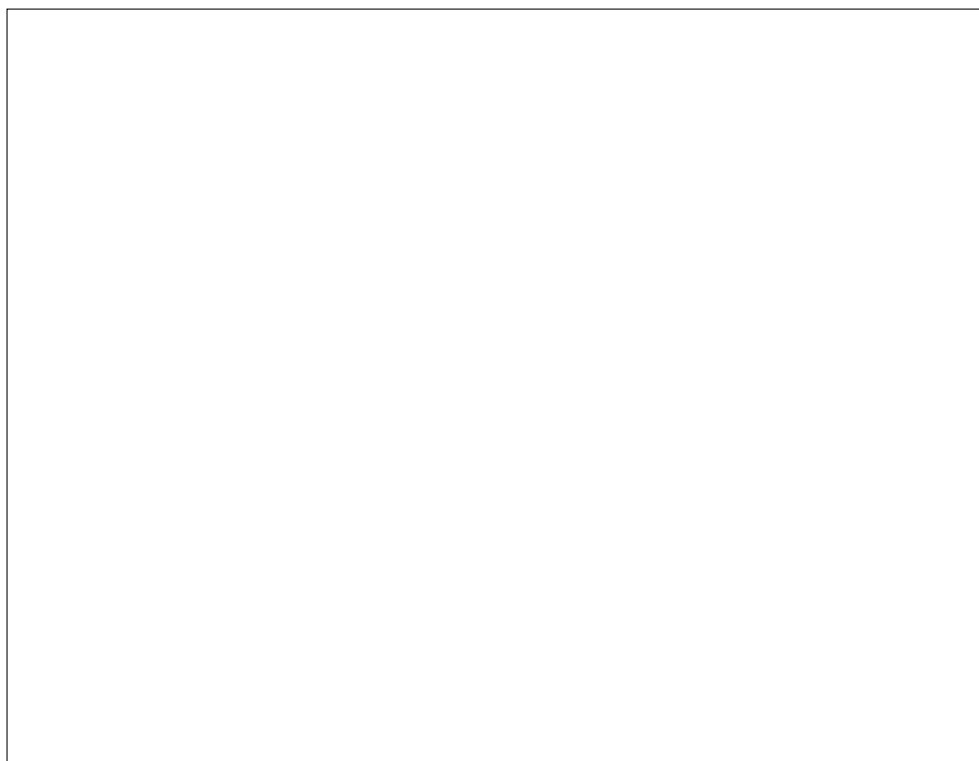
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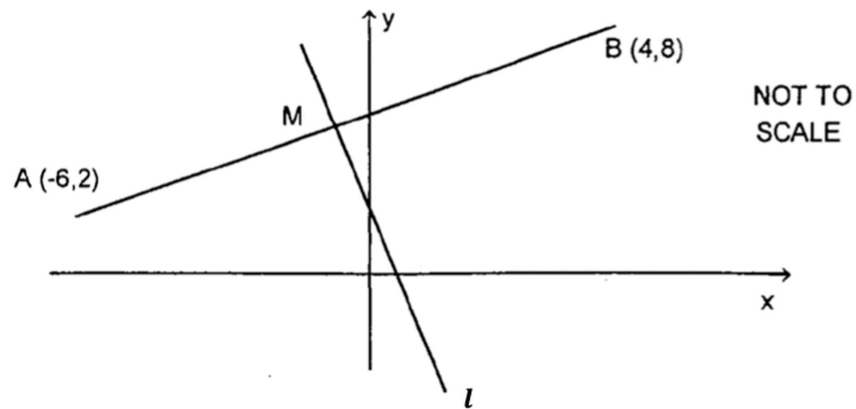
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iii. Sketch the graph of $y = f(x)$ **3**



- (f) The diagram below show $A(-6, 2)$ and $B(4, 8)$.



- i. Find M , the midpoint of AB . 1

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- ii. Show that the gradient of AB is $\frac{3}{5}$. 1

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- iii. Line l is the perpendicular bisector of AB . Show that the equation of line l is $5x + 3y - 10 = 0$. 2

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Question 12 (10 Marks)**Marks**

(a) Find the derivative of the followings, leaving your answer in simplest form.

i. $y = -3x^4$ **1**

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ii. $f(x) = (3x^2 - 5)^6$ **2**

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iii. $f(x) = \frac{2x - 1}{x + 4}$ **2**

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iv. $y = 3x^2\sqrt{x-2}$. (No need to fully simplify) **2**

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(b) Differentiate $f(x) = x^2$ from first principles. **2**

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- (c) Find the x coordinate of the point on the curve $y = x^2 + 5$ where the tangent has the gradient of -2 .

1

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Examination continues overleaf...

Question 13 (16 Marks)**Marks**

- (a) Show that $f(x) = \frac{1}{x^3} - x^3$ is always decreasing. **2**

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- (b) Consider $f(x) = x^3 + x^2 - x - 1$.
- i. Given that $x^3 + x^2 - x - 1 = (x - 1)(x + 1)^2$, find the x and y -intercepts. **2**

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- ii. Find the stationary points and determine their nature. **4**

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- iii. Find the point of inflexion.

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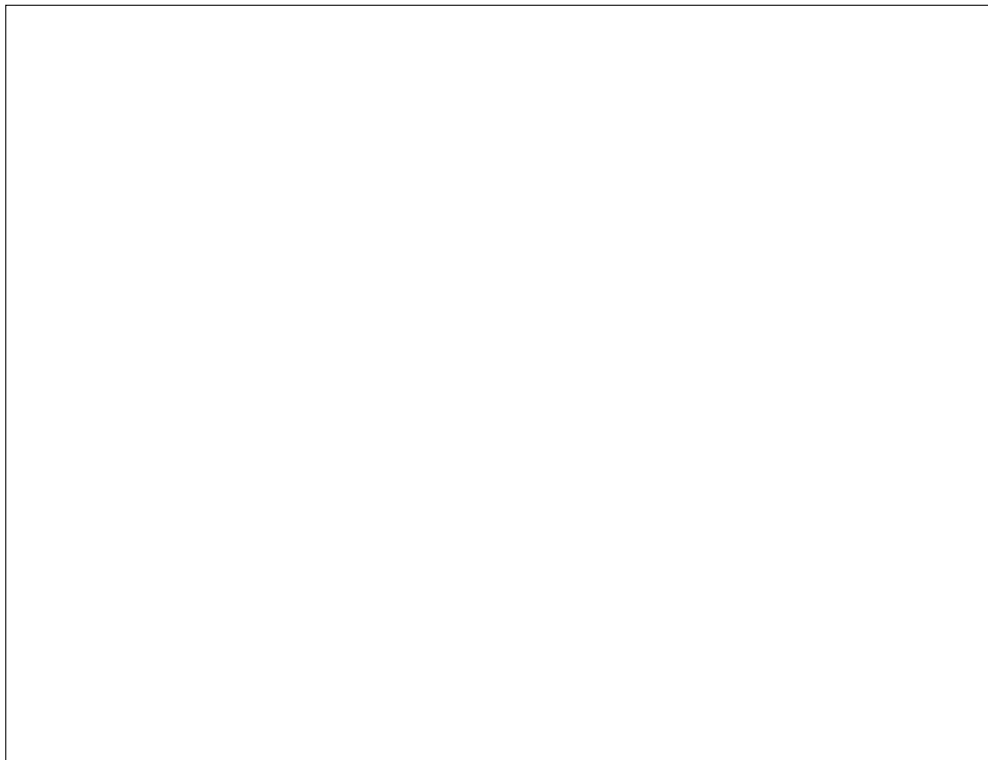
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- iv. Sketch $y = x^3 + x^2 - x - 1$, showing all important features.

2

- v. How many solutions does the equation $x^3 + x^2 - x - 1 = 1$ have?

1

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- (c) The volume, in litres, of water in a bathtub, t minutes after it has started to drain, is given by the equation

$$V = 50(14 - t)^2$$

- i. Calculate the initial volume of water in the bathtub. **1**

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- ii. At what rate does the water drain out of the bathtub after 6 minutes? **2**

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Examination continues overleaf...

Question 14 (16 Marks)**Marks**

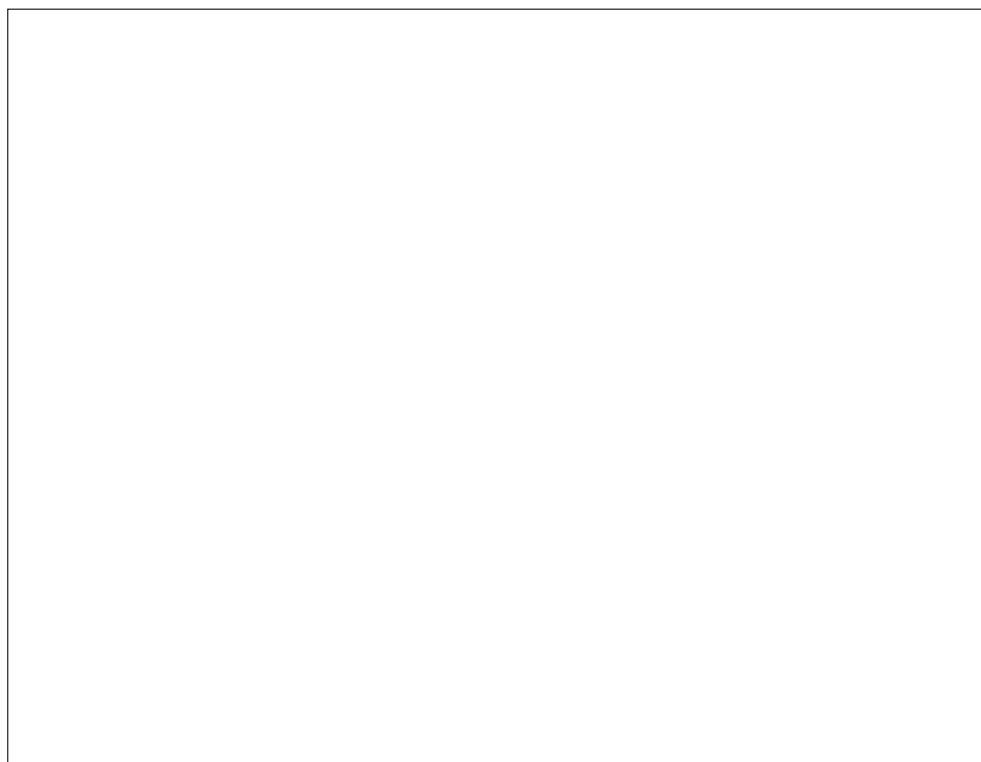
- (a) The angle of a sector in a circle of radius 9 cm is $\frac{\pi}{5}$ radians. Find the exact value of the perimeter of the sector. **2**

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- (b) i. On the same number plane, graph $y = \cos x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$, showing all intercepts. **2**



- ii. Show that the curves first intersect at $x = \frac{\pi}{6}$. **1**

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- (c) Show that the exact value of $\cos \frac{\pi}{4} + \sin \frac{2\pi}{3}$ is $\frac{\sqrt{2} + \sqrt{3}}{2}$ **1**

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- (d) i. Show that

2

$$\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$$

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- ii. Hence, or otherwise, solve

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$$\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$$

where $0 \leq x \leq 2\pi$.

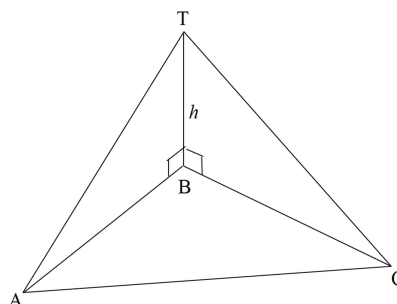
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- (e) The angle of elevation of a vertical tower BT of height h metres, at a point A due west of it, is 14° . From the point C , the bearing of the tower is 343° and the angle of elevation is 11° . The points A and C are 1 000 m apart and on the same level as the base B of the tower.



- i. By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$. 1

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- ii. Show that $AB = h \tan 76^\circ$ 1

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- iii. Find a similar expression for CB . 1

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- iv. Show that 2

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

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- v. Hence, find the height of the tower, correct to the nearest metre. 1

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Examination continues overleaf...

Question 15 (13 Marks)**Marks**

- (a) Solve the equation
- $2 \log_e(x - 5) = \log_e(2x - 7)$

3

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- (b) Differentiate with respect to
- x
- :

i. $y = e^{7x}$

1

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ii. $y = \frac{e^x}{x^2}$

2

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- (c) Find the equation of the normal to the curve
- $y = e^x(x^2 + 1)$
- at
- $(0, 1)$
- .

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- (d) i. Solve for x : $x^2 + 2x + 1 = 0$. **1**

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- ii. The displacement of an object at time (t) seconds is given by **3**

$$x = e^{-2t} + 4e^{-t} - 2t$$

Hence, or otherwise, show that the object never comes to rest.
(Hint: Let $m = e^{-t}$)

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End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

NESA STUDENT #:

Class (please ✓)

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)



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NAME

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Section I

10 marks

Attempt Question 1 to 10

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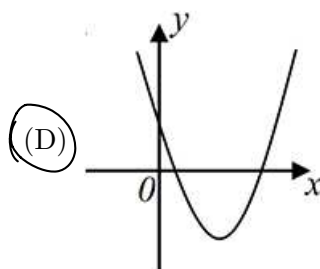
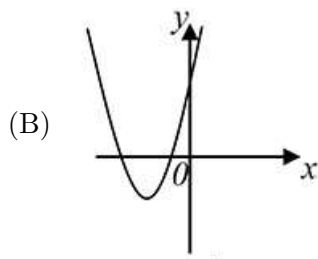
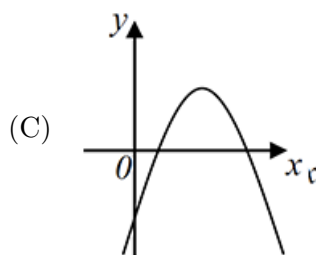
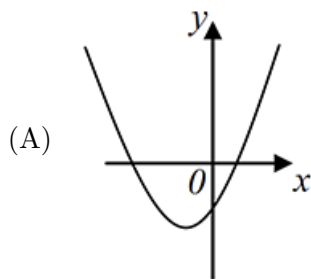
Questions

Marks

1. When the denominator is rationalised, what is $\frac{1}{\sqrt{5} - \sqrt{3}}$ equal to? 1

(A) $\frac{\sqrt{5} - \sqrt{3}}{2}$ (B) $\frac{\sqrt{5} - \sqrt{3}}{16}$ (C) $\frac{\sqrt{5} + \sqrt{3}}{2}$ (D) $\sqrt{5} + \sqrt{3}$

2. Which of the following parabolas could have the equation $y = (x - 5)(x - 1)$? 1



3. What is range of the function $f(x) = \sqrt{4 - x^2}$? 1

(A) $(0, 4)$ (B) $[-4, 4]$ (C) $(-2, 2)$ (D) $[0, 2]$

4. What is $e^{2 \log_e x}$ is equal to? 1

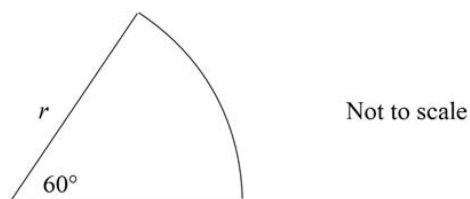
(A) $\sqrt{x}$ (B) x^2 (C) $2x$ (D) $e^2 x$

5. If $4^{x-1} = 32$, what is the value of x ? 1

(A) 10 (B) 3.5 (C) 3 (D) 6

6. The sector below has an area of 20π square units.

1

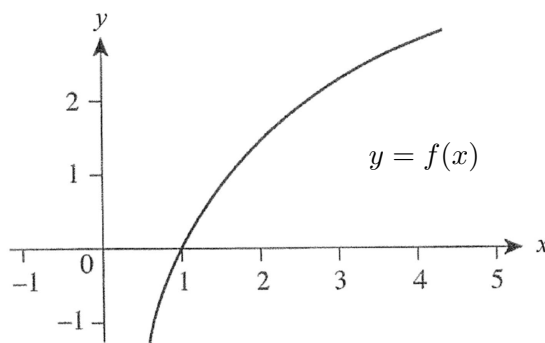


What is the value of r ?

- (A) $\sqrt{120}$ (B) $\sqrt{120\pi}$ (C) $\sqrt{\frac{\pi}{3}}$ (D) $\sqrt{\frac{1}{3}}$

7. The graph of $y = f(x)$ below has which of the following properties?

1



- (A) $f'(x) > 0$ and $f''(x) > 0$ (C) $f'(x) < 0$ and $f''(x) > 0$

- (B) $f'(x) > 0$ and $f''(x) < 0$ (D) $f'(x) < 0$ and $f''(x) < 0$

8. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$

1

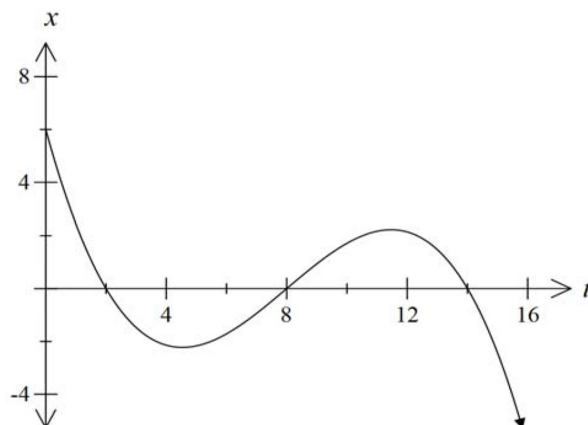
- (A) ∞ (B) 0 (C) 4 (D) 2

9. If $f(x) = a^x + a^{-x}$, which of the following expressions is equal to $[f(x)]^2$?

1

- (A) $1 - f(2x)$ (B) $1 + f(2x)$ (C) $2 - f(2x)$ (D) $2 + f(2x)$

10. The displacement, x metres, from the origin of a particle moving in a straight line at any time (t seconds) is shown in the graph. 1



When was the particle first at rest?

- (A) $t = 0$ (B) $t = 2$ (C) $t = 4.5$ (D) Never

Examination continues overleaf...

Section II

70 marks

Attempt Questions 11 to 15

Allow approximately 1 hour and 50 minutes for this section.

Question 11 (18 Marks)

Marks

- (a) Solve for x : $\frac{x+5}{3} - 2x = 7$ 2

$$x+5-6x=21$$

$$-5x=16$$

$$\therefore x = \frac{-16}{5}$$

- (b) Factorise $3x^2 + 5x - 2$. 2

$$\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}$$

$$= (x+2)(3x-1)$$

- (c) Simplify $\frac{a^2 \times a^{x-4}}{a^{1-2x}}$ 2

$$= \frac{a^{x-2}}{a^{1-2x}}$$

$$= a^{x-2-(1-2x)}$$

$$= a^{x-2-1+2x}$$

$$= a^{3x-3}$$

- (d) Consider the function $f(x) = \frac{3}{x+1}$. 1

- i. Show that $f(x)$ is neither an even nor an odd function. 1

$$f(-x) = \frac{3}{-x+1}$$

$$= -\frac{3}{x-1}$$

$$\neq -f(x) \quad \therefore \text{neither even nor odd}$$

- ii. State the domain of $y = f(x)$ in interval notation. 1

$$x \in (-\infty, -1) \cup (-1, \infty)$$

(e) Given

$$f(x) = \begin{cases} -4 & x \leq -2 \\ x & -2 < x \leq 1 \\ x^2 - 1 & x > 1 \end{cases}$$

i. Find $f(-1)$.

1

$$f(-1) = -1$$

ii. Find $f(a^2 + 1)$. Simplify your answer.

2

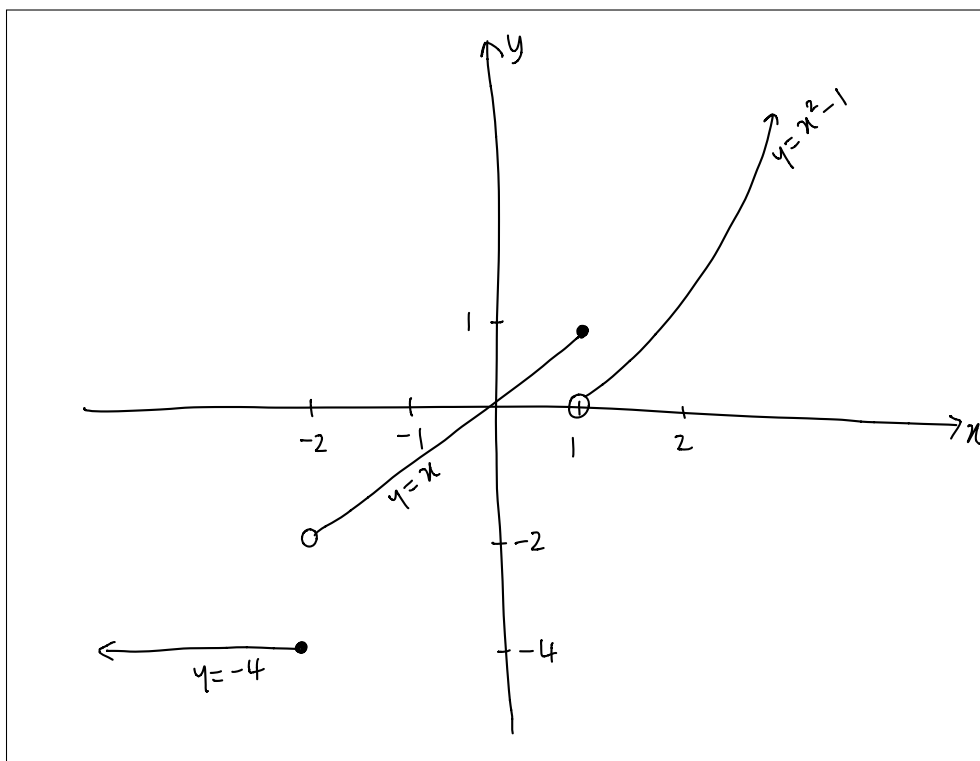
$$f(a^2 + 1) = (a^2 + 1)^2 - 1$$

$$= a^4 + 2a^2 + 1 - 1$$

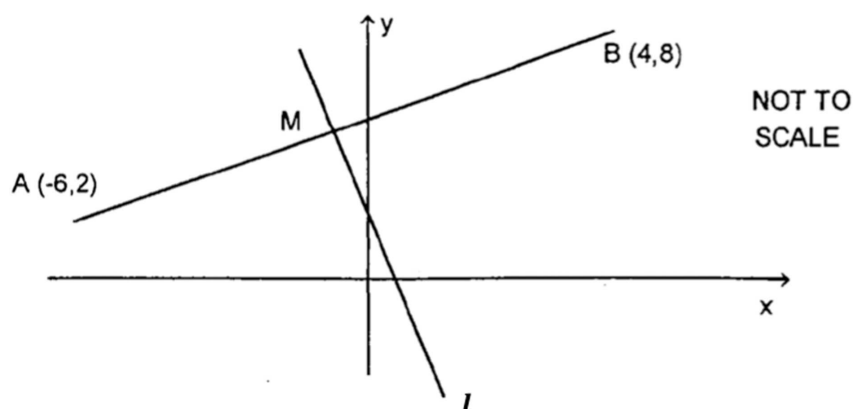
$$= a^4 + 2a^2$$

iii. Sketch the graph of $y = f(x)$

3



- (f) The diagram below show $A(-6, 2)$ and $B(4, 8)$.



- i. Find M , the midpoint of AB .

1

$$M\left(\frac{-2}{2}, \frac{10}{2}\right)$$

$$= M(-1, 5)$$

- ii. Show that the gradient of AB is $\frac{3}{5}$.

1

$$m_{AB} = \frac{8-2}{4-(-6)}$$

$$= \frac{6}{10} = \frac{3}{5}$$

- iii. Line l is the perpendicular bisector of AB . Show that the equation of line l is $5x + 3y - 10 = 0$.

2

$$m_l = -\frac{5}{3}, \quad (-1, 5)$$

$$\therefore \text{Eqn: } y - 5 = -\frac{5}{3}(x + 1)$$

$$3y - 15 = -5x - 5$$

$$5x + 3y - 10 = 0$$

Question 12 (10 Marks)**Marks**

(a) Find the derivative of the followings, leaving your answer in simplest form.

i. $y = -3x^4$ 1

$$y' = -12x^3$$

ii. $f(x) = (3x^2 - 5)^6$ 2

$$f'(x) = 6(3x^2 - 5)^5 \cdot 6x$$

$$= 36x(3x^2 - 5)^5$$

iii. $f(x) = \frac{2x - 1}{x + 4}$ 2

$$f'(x) = \frac{2(x+4) - (2x-1)}{(x+4)^2}$$

$$= \frac{2x+8-2x+1}{(x+4)^2}$$

$$= \frac{9}{(x+4)^2}$$

iv. $y = 3x^2\sqrt{x-2}$. (No need to fully simplify) 2

$$= 3x^2(x-2)^{1/2}$$

$$y' = 6x(x-2)^{1/2} + 3x^2 \cdot \frac{1}{2}(x-2)^{-1/2}$$

$$= 6x\sqrt{x-2} + \frac{3x^2}{2\sqrt{x-2}}$$

(b) Differentiate $f(x) = x^2$ from first principles. 2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h$$

$$= 2x$$

- (c) Find the x coordinate of the point on the curve $y = x^2 + 5$ where the tangent has the gradient of -2 .

1

$$y' = 2x$$

$$\therefore 2x = -2$$

$$x = -1$$

Examination continues overleaf...

Question 13 (16 Marks)**Marks**

- (a) Show that $f(x) = \frac{1}{x^3} - x^3$ is always decreasing. 2

$$f(x) = x^{-3} - x^3$$

$$f'(x) = -3x^{-4} - 3x^2$$

$$= -\frac{3}{x^4} - 3x^2$$

For all real x , $x^2 \geq 0$ & $x^4 \geq 0$

$$\therefore -\frac{3}{x^4} - 3x^2 < 0 \text{ for all } x.$$

$\therefore$ always decreasing

- (b) Consider $f(x) = x^3 + x^2 - x - 1$. 2
- i. Given that $x^3 + x^2 - x - 1 = (x-1)(x+1)^2$, find the x and y -intercepts.

$$x\text{-int} : x = 1, -1$$

$$y\text{-int} : y = -1$$

- ii. Find the stationary points and determine their nature. 4

$$f'(x) = 3x^2 + 2x - 1$$

$$\frac{1}{3} \quad -1$$

$$= (x+1)(3x-1)$$

$$\therefore x = -1, \frac{1}{3}$$

$$y = 0, -\frac{32}{27} \quad \therefore (-1, 0) \text{ \& \; } (\frac{1}{3}, -\frac{32}{27})$$

x	-2	-1	0	$\frac{1}{3}$	1
y'	7	0	-1	0	4
	/	—	\	—	/

$\therefore$ max @ $(-1, 0)$

min @ $(\frac{1}{3}, -\frac{32}{27})$

- iii. Find the point of inflexion.

2

$$f''(x) = 6x + 2$$

$$6x + 2 = 0$$

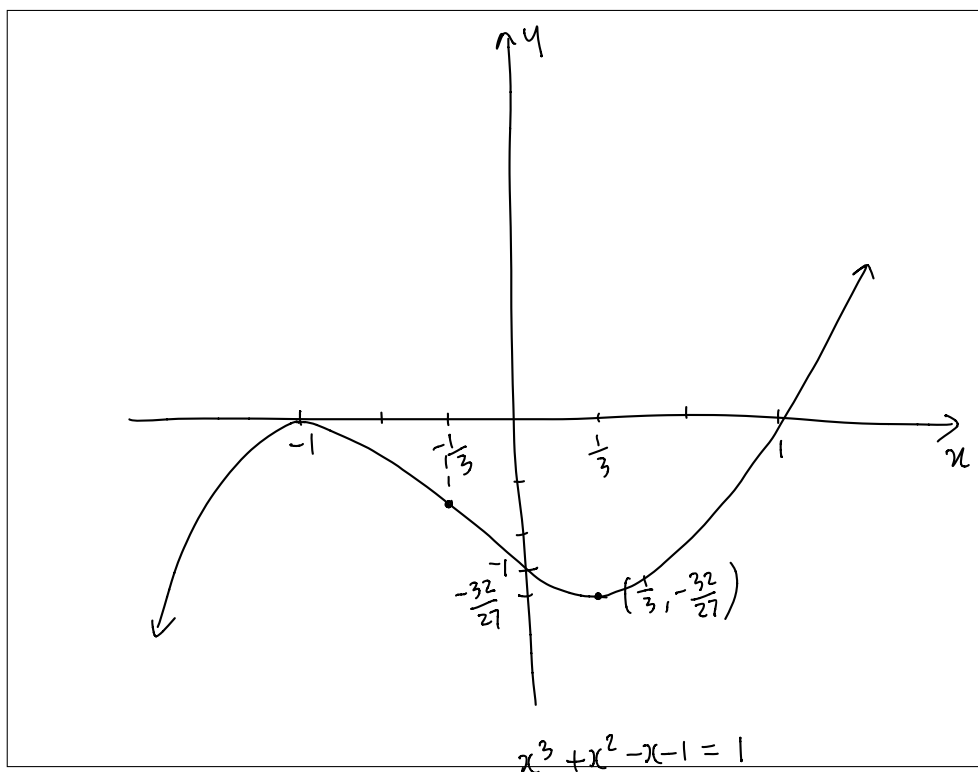
$$x = -\frac{1}{3}, \quad y = -\frac{16}{27}$$

x	-1	$-\frac{1}{3}$	0
y''	-4	0	2

$\therefore$ pt of inflexion @ $(-\frac{1}{3}, -\frac{16}{27})$

- iv. Sketch
- $y = x^3 + x^2 - x - 1$
- , showing all important features.

2



- v. How many solutions does the equation
- $x^3 + x^2 - x - 1 = 0$
- have?

1

1 solution

- (c) The volume, in litres, of water in a bathtub, t minutes after it has started to drain, is given by the equation

$$V = 50(14 - t)^2$$

- i. Calculate the initial volume of water in the bathtub. **1**

When $t=0$, $V = 50 \times 14^2$
 $= 9800 \text{ L}$

- ii. At what rate does the water drain out of the bathtub after 6 minutes? **2**

$\frac{dV}{dt} = 100(14-t)^1 \times (-1)$
 $= -100(14-t)$

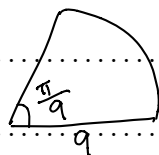
When $t=6$, $\frac{dV}{dt} = -800 \text{ L/min}$

$\therefore$ The water drains out at 800 L/min .

Examination continues overleaf...

Question 14 (16 Marks)**Marks**

- (a) The angle of a sector in a circle of radius 9 cm is $\frac{\pi}{5}$ radians. Find the exact value of the perimeter of the sector. **2**



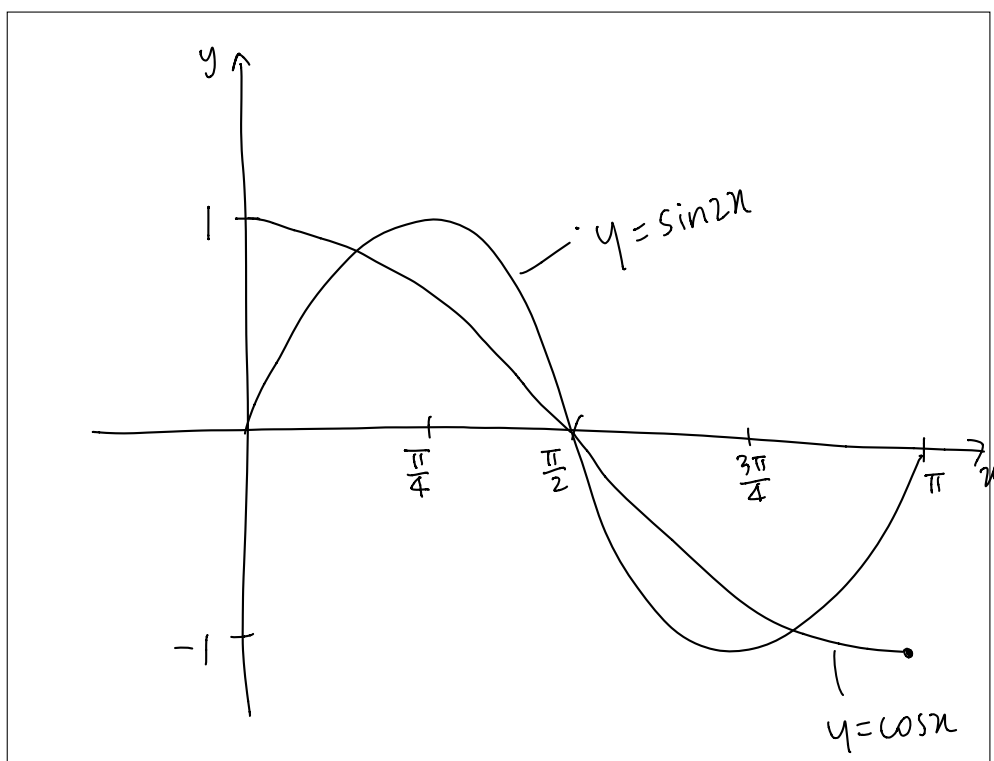
$$l = r\theta = 9 \times \frac{\pi}{5} = \frac{9\pi}{5}$$

$$\therefore \left(\frac{9\pi}{5} + 18 \right) \text{ cm}$$

- (b) i. On the same number plane, graph $y = \cos x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$, showing all intercepts. **2**

$$T = 2\pi$$

$$T = \pi$$



- ii. Show that the curves first intersect at $x = \frac{\pi}{6}$. **1**

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin 2\left(\frac{\pi}{6}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \quad \therefore \text{Intersect @ } x = \frac{\pi}{6}$$

- (c) Show that the exact value of $\cos \frac{\pi}{4} + \sin \frac{2\pi}{3}$ is $\frac{\sqrt{2} + \sqrt{3}}{2}$ **1**

$$\cos \frac{\pi}{4} + \sin \frac{2\pi}{3} = \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{2} + \sqrt{3}}{2}$$

- (d) i. Show that

2

$$\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$$

$$\text{LHS} = \frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta$$

$$= \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$$

$$= \tan \theta$$

$$= \text{RHS}$$

- ii. Hence, or otherwise, solve

2

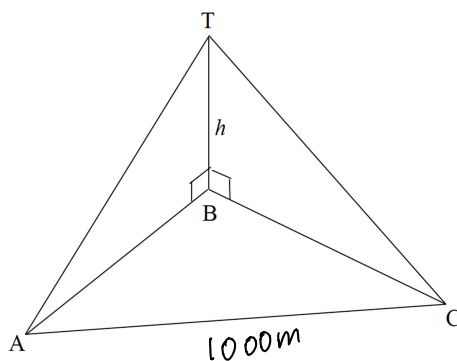
$$\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$$

where $0 \leq \theta \leq 2\pi$.

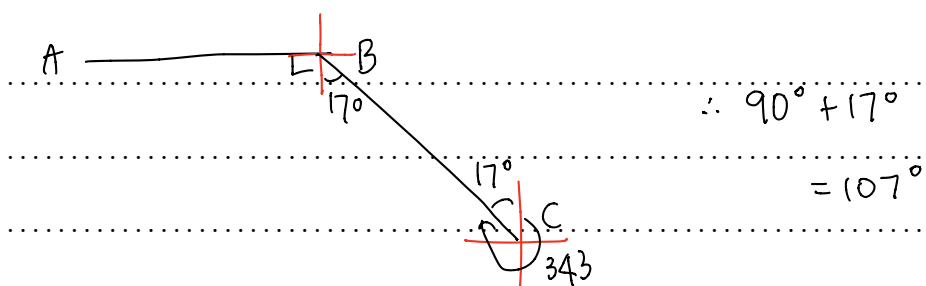
$$\tan \theta - \sin \theta \cos \theta = 1 - \sin \theta \cos \theta$$

$$\tan \theta = 1, \quad \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

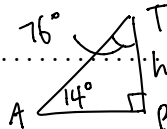
- (e) The angle of elevation of a vertical tower
- BT
- of height
- h
- metres, at a point
- A
- due west of it, is
- 14°
- . From the point
- C
- , the bearing of the tower is
- 343°
- and the angle of elevation is
- 11°
- . The points
- A
- and
- C
- are 1 000 m apart and on the same level as the base
- B
- of the tower.



- i. By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$. 1



- ii. Show that $AB = h \tan 76^\circ$ 1



$$\tan 76^\circ = \frac{AB}{h}$$

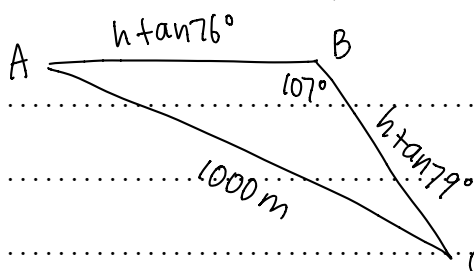
$$\therefore AB = h \tan 76^\circ$$

- iii. Find a similar expression for CB . 1

$$CB = h \tan 79^\circ$$

- iv. Show that 2

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$



cosine rule

$$1000^2 = h^2 \tan^2 76^\circ + h^2 \tan^2 79^\circ - 2 \times h \tan 76^\circ \times h \tan 79^\circ \times \cos 107^\circ$$

$$1000^2 = h^2 (\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ)$$

$$\therefore h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

- v. Hence, find the height of the tower, correct to the nearest metre. 1

$$h = 135.310 \dots = 135 \text{ m (to the nearest m)}$$

Examination continues overleaf...

Question 15 (13 Marks)**Marks**

- (a) Solve the equation $2\log_e(x-5) = \log_e(2x-7)$ **3**

$$(x-5)^2 = 2x-7$$

$$x^2 - 10x + 25 = 2x - 7$$

$$x^2 - 12x + 32 = 0$$

$$(x-4)(x-8) = 0$$

$$x = 4, 8$$

As $x-5 > 0$, $x=8$ is the only solution.

- (b) Differentiate with respect to x :

i. $y = e^{7x}$

$$y' = 7e^{7x}$$

ii. $y = \frac{e^x}{x^2}$

$$y' = \frac{e^x \cdot x^2 - 2xe^x}{x^4}$$

$$= \frac{xe^x - 2e^x}{x^3}$$

$$= \frac{e^x(x-2)}{x^3}$$

- (c) Find the equation of the normal to the curve $y = e^x(x^2 + 1)$ at $(0, 1)$. **3**

$$y' = e^x(2x) + e^x(x^2 + 1)$$

$$= e^x(x^2 + 2x + 1)$$

$$= e^x(x+1)^2$$

At $(0, 1)$, $m_{tan} = 1$, $\therefore m_{nor} = -1$

$\therefore$ Eqn of normal is

$$y - 1 = -x$$

$$y = -x + 1$$

- (d) i. Solve for x : $x^2 + 2x + 1 = 0$. 1

$$(x+1)^2 = 0$$

$$x = -1$$

- ii. The displacement of an object at time (t) seconds is given by 3

$$x = e^{-2t} + 4e^{-t} - 2t$$

show that ~~find the time(s)~~ never

Hence, or otherwise, ~~find the time(s)~~ the object comes to rest. (Hint: Let $m = e^{-t}$)

$$\dot{x} = -2e^{-2t} - 4e^{-t} - 2$$

$$= -2(e^{-2t} + 2e^{-t} + 1)$$

$$= -2(m^2 + 2m + 1)$$

$$= -2(m+1)^2$$

$$\text{when at rest, } m = -1$$

$$\therefore e^{-t} = -1$$

no solutions

$\therefore$ never come to rest.

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

NESA STUDENT #:

Class (please ✓)

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)